

Literature status: Problem 2 on asymptotically symmetric spaces

Run `fa_banach_001`; source arXiv:math/0508035

Source problem

Junge, Kutzarova, and Odell introduced asymptotically symmetric Banach spaces and asked:

If X is infinite dimensional and asymptotically symmetric, must X contain an asymptotic ℓ_p subspace for some p ?

This is Problem 2 in arXiv:math/0508035.

Later literature

Kutzarova and Motakis later wrote *Asymptotically symmetric spaces with hereditarily non-unique spreading models*, arXiv:1902.10098. In the introduction they cite the Junge–Kutzarova–Odell problem in the slightly stronger form asking whether every asymptotically symmetric Banach space contains an infinite-dimensional asymptotic- ℓ_p or asymptotic- c_0 subspace. They state that they solve this problem in the negative direction by a variation $\mathfrak{X}_{0,1}^{1/2}$ of the Argyros–Beanland–Motakis space $\mathfrak{X}_{0,1}$.

The literature answer

Theorem 1. *Problem 2 of arXiv:math/0508035 has a negative answer.*

Proof. Let $X = \mathfrak{X}_{0,1}^{1/2}$ be the space constructed in arXiv:1902.10098. The conclusion section of that paper proves that X is asymptotically symmetric.

The same paper proves that every block subspace of X admits both a normalized block sequence generating an ℓ_1 spreading model and a normalized block sequence generating a c_0 spreading model. It also recalls the standard obstruction: if a Banach space is asymptotic- ℓ_p or asymptotic- c_0 , then every spreading model generated by a normalized weakly null sequence in that space must be equivalent to the corresponding unit vector basis.

Therefore no infinite-dimensional subspace of X can be asymptotic- ℓ_p for any $1 \leq p < \infty$. If such a subspace were asymptotic- ℓ_p , its weakly-null spreading models would all have to be ℓ_p -type; but the Kutzarova–Motakis hereditary property supplies both ℓ_1 and c_0 spreading models in every subspace. The same argument also rules out asymptotic- c_0 subspaces.

Thus X is an infinite-dimensional asymptotically symmetric Banach space with no asymptotic- ℓ_p subspace. This is exactly a counterexample to the conclusion asked for in the 2005 problem. $\square$

Verification notes

The implication used above is stronger than the original problem requires: arXiv:1902.10098 rules out asymptotic- ℓ_p and asymptotic- c_0 subspaces, while the source problem asks only for an asymptotic- ℓ_p subspace. The supporting paper also proves reflexivity of $\mathfrak{X}_{0,1}^{1/2}$, so there is no finite-dimensional or nonseparable edge case involved.

This packet is a literature-status identification, not a new proof. It answers Problem 2 of arXiv:math/0508035. It does not address Problem 1 of the same source paper on infinite-dimensional

subspaces of noncommutative $L_p(N)$, nor the separate final question in arXiv:1902.10098 about whether every asymptotically symmetric Banach space admits an ℓ_p or c_0 spreading model.

The run lightweight indexes were searched for arXiv:0508035, the source title, “asymptotically symmetric”, “spreading model”, and the author names before promotion. No existing exact packet or attempt for this problem was found.

Human review recommendation

Treat Junge–Kutzarova–Odell Problem 2 as already answered negatively by Kutzarova–Motakis arXiv:1902.10098.

References

- [1] M. Junge, D. Kutzarova, and E. Odell, *On asymptotically symmetric Banach spaces*, Studia Math. 173 (2006), no. 3, 203–231; arXiv:math/0508035.
- [2] D. Kutzarova and P. Motakis, *Asymptotically symmetric spaces with hereditarily non-unique spreading models*, arXiv:1902.10098.